

Optimization

Advanced Statistics Lab

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Formal Definition

Choose values that make a function small or large.

We start with an ordinary function.

Least squares and likelihood come later.

Basic Optimization Problem



OBJECTIVE

Let θ be the parameter vector and Θ the allowed parameter space.

$$\hat{\theta} \in \arg \min_{\theta \in \Theta} f(\theta)$$

For maximization:

$$\arg \max_{\theta \in \Theta} f(\theta) = \arg \min_{\theta \in \Theta} -f(\theta)$$

Symbol Map

Symbol	Meaning
θ	Candidate value or parameter vector.
Θ	Allowed set of candidate values.
$\hat{\theta}$	Optimizing value, marked with a hat because it has been selected or estimated.
$f(\theta)$	Objective function.
$\arg \min$	Input value that gives the minimum.
$\arg \max$	Input value that gives the maximum.

Constrained Problems

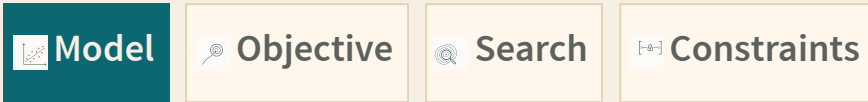
CONSTRAINTS

$$\begin{aligned} \hat{\theta} &\in \arg \min_{\theta \in \Theta} f(\theta) \\ \text{subject to } g_j(\theta) &\leq 0, \quad j = 1, \dots, m \\ h_k(\theta) &= 0, \quad k = 1, \dots, r \end{aligned}$$

Constraints narrow the parameter space.

Symbol	Meaning
$g_j(\theta)$	The j th inequality constraint.
$h_k(\theta)$	The k th equality constraint.
m	Number of inequality constraints.
r	Number of equality constraints.

The Workflow



1. Define the model.
2. Define the objective.
3. Search for good parameter values.
4. Check whether the result makes statistical and biological sense.

First Example: Fence A Paddock



MODEL

80 meters of fence.

$$2w + 2\ell = 80$$

Divide by 2:

$$w + \ell = 40$$

So:

$$\ell(w) = 40 - w$$

$$A(w) = w(40 - w)$$

Area Function



OBJECTIVE

```
fence_m <- 80

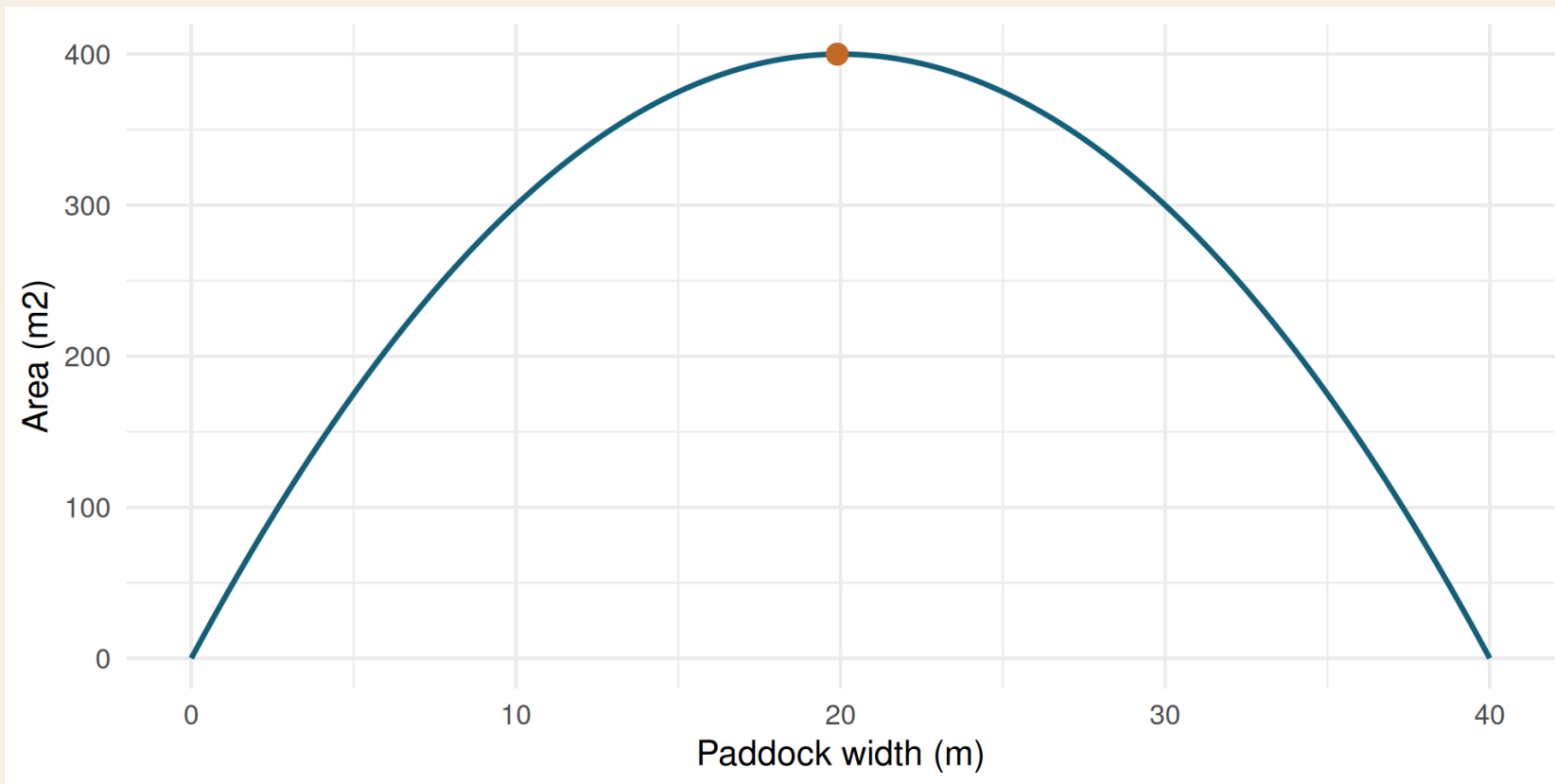
paddock_area <- function(width, fence_m = 80) {
  length <- fence_m / 2 - width
  width * length
}

candidate_widths <- seq(0, fence_m / 2, length.out = 200)
area_grid <- data.frame(width = candidate_widths)
area_grid$area <- paddock_area(area_grid$width, fence_m)
best_grid <- area_grid[which.max(area_grid$area), ]
```

$$\hat{w} = \arg \max_{0 \leq w \leq 40} A(w)$$

Objective Shape

```
ggplot(area_grid, aes(width, area)) +  
  geom_line(color = "#155f7a", linewidth = 1.2) +  
  geom_point(data = best_grid, color = "#c46a2b", size = 4) +  
  labs(x = "Paddock width (m)", y = "Area (m2)")
```



Search With `optim()`



SEARCH

Maximize area by minimizing negative area.

```
negative_area <- function(width, fence_m = 80) {  
  -paddock_area(width, fence_m)  
}  
  
fit_area <- optim(  
  par = 10,  
  fn = negative_area,  
  fence_m = fence_m,  
  method = "L-BFGS-B",  
  lower = 0,  
  upper = fence_m / 2  
)  
  
data.frame(width = fit_area$par, area = -fit_area$value)
```

	width	area
1	20	400

Bound Constraints

CONSTRAINTS

$$0 \leq w \leq 40$$

```
fit_tight_area <- optim(  
  par = 10,  
  fn = negative_area,  
  fence_m = fence_m,  
  method = "L-BFGS-B",  
  lower = 0,  
  upper = 15  
)  
  
data.frame(  
  problem = c("correct bounds", "too-tight upper bound"),  
  width = c(fit_area$par, fit_tight_area$par),  
  area = c(-fit_area$value, -fit_tight_area$value)  
)
```

	problem	width	area
1	correct bounds	20	400
2	too-tight upper bound	15	375

Where This Goes Next

 MODEL

Same workflow, statistical objectives:

Topic	Objective
Least squares	Minimize squared vertical distances.
Maximum likelihood	Maximize probability of the observed data.
Animal models	Balance fit and shrinkage.

Exercises

1. Change the amount of fencing.
2. Change the starting value in `optim()`.
3. Make the upper bound too tight.
4. Rewrite the objective for a three-sided fence.